

Phase 1: Data Science Project Proposals

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Project 1: Housing Amenities Impact on Apartment Listing Prices

Business Issue and Overview

Real estate investors and property managers lack quantitative evidence on which amenities drive rental pricing. Understanding amenity importance enables strategic property improvements and competitive pricing. This project analyzes which amenities (parking, laundry, pet policies, fitness centers, etc.) most significantly influence rental prices across U.S. markets.

Business Questions

- Which amenities have strongest impact on rental pricing?
- How does amenity importance vary across geographic regions?
- What ROI can property managers expect from amenity investments?
- How do amenity combinations affect pricing?

Data Needs

Major entities: Rental listings (price, location, bedrooms, bathrooms, square footage), amenities (binary/categorical flags), geographic clusters (region, state), listing metadata (date listed, availability).

Motivation

Real-world business application with immediate utility. Property investment decisions rest on amenity impact analysis. Combines regression modeling with feature importance interpretation—core data science competencies.

Project 2: Student Course Success Prediction

Business Issue and Overview

Educational institutions face resource constraints and must identify at-risk students early. This project builds a predictive model to identify students likely to struggle or fail in courses,

enabling early intervention programs. Predictions target first-semester students to maximize intervention effectiveness.

Business Questions

- Which student characteristics predict course success?
- When in the semester should interventions occur?
- What baseline GPA threshold indicates risk?
- How accurately can we identify at-risk students?

Data Needs

Major entities: Student demographics (age, prior GPA, major, course history), course characteristics (difficulty, subject, instructor), performance metrics (grades, attendance, engagement).

Motivation

Addresses education sector challenges with 7-week feasibility. Classification modeling (success/at-risk) is computationally simpler than regression. Public datasets available (student performance datasets). Focuses on interpretability—why students succeed—valuable for stakeholder communication.

Recommended Project

Project 1 (Housing Amenities) is recommended. It aligns with your prior work, has established data availability (Craigslist listings), and delivers concrete business value for real estate stakeholders. The feature importance analysis directly answers investor questions about amenity ROI.